

Model Lab

Every prior, same slate — 2026-10-03

SERVED PRIOR

Served (full season)

PRIORS COMPARED

11

HELD-OUT PA

35,167

Held-Out Scores

Fitted on plate appearances before 2026-08-24, scored on the 35,167 that came after (actual HR rate 3.03%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5869	1.55×	0.02931	0.13477	0.00535
KNN comps Prior is the pooled rate of the nearest comparables.	0.5975	1.52×	0.02928	0.13399	0.00444
SVR Support-vector regression on the same feature matrix.	0.5959	1.52×	0.02933	0.13473	0.00620
Random forest Forest regression; a learned similarity metric.	0.5948	1.50×	0.02931	0.13431	0.00587
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5975	1.52×	0.02928	0.13399	0.00444
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5975	1.52×	0.02928	0.13399	0.00444
Linear regression Ordinary least squares on the same features.	0.5965	1.52×	0.02930	0.13430	0.00426
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5964	1.52×	0.02932	0.13441	0.00649
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5966	1.53×	0.02930	0.13423	0.00541
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5958	1.49×	0.02932	0.13439	0.00609

Blend weights (forest variant): KNN 1.00, SVR 0.00, RF 0.00

Shrinkage K per prior: league 255, knn 255, svr 255, rf 255, core 255, ensemble 255, linear 255, logistic 255, nn 255, xgb 255 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,884 PA, actual HR rate 2.37%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5546	1.31×	0.02316	0.11258	0.01010
KNN comps Prior is the pooled rate of the nearest comparables.	0.5617	1.20×	0.02324	0.11346	0.01486
SVR Support-vector regression on the same feature matrix.	0.5766	1.20×	0.02351	0.11826	0.02104
Random forest Forest regression; a learned similarity metric.	0.5854	1.20×	0.02342	0.11473	0.01913
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5617	1.20×	0.02324	0.11346	0.01486
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5617	1.20×	0.02324	0.11346	0.01486
Linear regression Ordinary least squares on the same features.	0.5716	1.20×	0.02336	0.11567	0.01697
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5757	1.09×	0.02342	0.11505	0.01775
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5796	1.31×	0.02331	0.11406	0.01505
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5924	1.09×	0.02351	0.11525	0.02230

Slate Agreement

Served (full season) · 6/6 shared 0/6 hit

Junior Caminero
Shohei Ohtani
Ben Rice
Matt Olson
Ronald Acuña Jr.
Will Smith

KNN comps · 6/6 shared 0/6 hit

Junior Caminero
Shohei Ohtani
Ben Rice
Ronald Acuña Jr.
Will Smith
Matt Olson

Random forest · 6/6 shared 0/6 hit

Junior Caminero
Shohei Ohtani
Ben Rice
Matt Olson
Will Smith
Ronald Acuña Jr.

Forest blend · 6/6 shared 0/6 hit

Junior Caminero
Shohei Ohtani
Ben Rice
Ronald Acuña Jr.
Will Smith
Matt Olson

Logistic regression · 6/6 shared 0/6 hit

Flat league · 5/6 shared 0/6 hit

19.5% Junior Caminero
19.2% Shohei Ohtani
18.5% Ben Rice
17.4% Will Smith
17.2% Ronald Acuña Jr.
17.0% Fernando Tatis Jr.

SVR · 6/6 shared 0/6 hit

19.4% Junior Caminero
18.6% Shohei Ohtani
18.3% Ben Rice
17.1% Matt Olson
17.1% Ronald Acuña Jr.
16.9% Will Smith

Core (KNN+SVR) · 6/6 shared 0/6 hit

19.9% Junior Caminero
19.2% Shohei Ohtani
18.6% Ben Rice
17.4% Ronald Acuña Jr.
17.3% Will Smith
17.1% Matt Olson

Linear regression · 6/6 shared 0/6 hit

19.4% Junior Caminero
18.6% Shohei Ohtani
18.3% Ben Rice
17.1% Will Smith
17.1% Matt Olson
16.9% Ronald Acuña Jr.

18.9%
17.3%
17.0%
16.6%
16.4%
16.0%

20.7%
19.1%
18.9%
17.7%
17.2%
16.6%

19.4%
18.6%
18.3%
17.1%
17.1%
16.9%

19.6%
18.9%
18.3%
17.6%
17.1%
17.0%

Junior Caminero	19.9%	Neural net (torch) · 6/6 shared	0/6 hit
Shohei Ohtani	19.3%	Junior Caminero	20.0%
Ben Rice	18.5%	Shohei Ohtani	19.2%
Matt Olson	17.8%	Ben Rice	18.5%
Will Smith	17.4%	Will Smith	17.5%
Ronald Acuña Jr.	17.1%	Matt Olson	17.5%
		Ronald Acuña Jr.	17.2%
		XGBoost · 6/6 shared	0/6 hit
Junior Caminero	20.4%		
Shohei Ohtani	19.6%		
Ben Rice	18.6%		
Matt Olson	17.5%		
Will Smith	17.0%		
Ronald Acuña Jr.	16.9%		

Hitter by Hitter

Hitter	Served (full season)	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression
Atlanta Braves · vs LHP · 42 HR / 711 PA	17.4%	15.8%	16.9%	17.7%	17.4%	16.9%	16.9%	17.1%	17.8%
5. Ronald Acuña Jr. Atlanta Braves · vs LHP · 20 HR / 482 PA	17.2%	16.4%	17.1%	17.2%	17.1%	17.1%	17.1%	17.0%	17.1%
6. Will Smith Los Angeles Dodgers · vs LHP · 10 HR / 276 PA	17.0%	16.6%	17.1%	16.6%	17.3%	17.1%	17.1%	17.6%	17.4%
7. Ryan Vilade Tampa Bay Rays · vs RHP · 11 HR / 278 PA	16.4%	15.4%	16.1%	15.7%	16.6%	16.1%	16.1%	15.8%	15.6%
8. Teoscar Hernández Los Angeles Dodgers · vs LHP · 16 HR / 491 PA	16.2%	15.5%	16.4%	16.3%	16.6%	16.4%	16.4%	16.5%	16.5%
9. Fernando Tatis Jr. slate San Diego Padres · vs RHP · 26 HR / 712 PA	16.1%	16.0%	16.4%	16.1%	15.9%	16.4%	16.4%	16.5%	16.7%
10. Victor Mesa Jr. Tampa Bay Rays · vs RHP · 14 HR / 262 PA	16.0%	15.6%	16.1%	16.3%	16.2%	16.1%	16.1%	16.1%	16.1%
11. Jackson Chourio Milwaukee Brewers · vs LHP · 25 HR / 573 PA	15.8%	15.9%	16.2%	16.5%	16.7%	16.2%	16.2%	16.4%	16.5%
12. Mookie Betts Los Angeles Dodgers · vs LHP · 23 HR / 528 PA	15.7%	15.7%	15.5%	16.3%	15.6%	15.5%	15.5%	15.9%	16.0%
13. Jackson Merrill San Diego Padres · vs RHP · 26 HR / 672 PA	15.6%	14.3%	15.4%	15.5%	15.8%	15.4%	15.4%	15.6%	16.0%
14. Jonathan Aranda Tampa Bay Rays · vs RHP · 22 HR / 668 PA	15.5%	14.7%	15.6%	15.1%	15.3%	15.6%	15.6%	15.4%	15.2%
15. Luis García Jr. New York Yankees · vs RHP · 31 HR / 549 PA	15.1%	14.5%	15.2%	15.6%	15.7%	15.2%	15.2%	15.4%	15.8%
16. Heliot Ramos New York Yankees · vs RHP · 13 HR / 445 PA	14.9%	14.4%	14.8%	14.3%	14.4%	14.8%	14.8%	15.0%	14.8%
17. Munetaka Murakami slate Chicago White Sox · vs LHP · 35 HR / 563 PA	14.8%	13.1%	14.8%	15.2%	15.0%	14.8%	14.8%	14.6%	14.7%
18. Drake Baldwin Atlanta Braves · vs LHP · 25 HR / 628 PA	14.8%	14.1%	14.9%	14.9%	15.3%	14.9%	14.9%	15.0%	14.9%

Hitter	Served (full season)	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression
19. Yandy Díaz Tampa Bay Rays · vs RHP · 22 HR / 686 PA	14.4%	14.7%	14.3%	14.8%	14.6%	14.3%	14.3%	14.7%	14.5%
20. Jake Bauers Milwaukee Brewers · vs LHP · 26 HR / 578 PA	14.1%	13.2%	14.2%	14.4%	14.4%	14.2%	14.2%	14.1%	14.1%
21. Garrett Mitchell Milwaukee Brewers · vs LHP · 11 HR / 531 PA	13.6%	13.1%	13.5%	12.6%	12.8%	13.5%	13.5%	13.0%	12.9%
22. Austin Riley Atlanta Braves · vs LHP · 20 HR / 643 PA	13.6%	12.3%	13.3%	13.1%	12.8%	13.3%	13.3%	13.1%	13.2%
23. Randal Grichuk Chicago White Sox · vs LHP · 15 HR / 283 PA	13.6%	12.3%	13.3%	14.2%	13.4%	13.3%	13.3%	13.3%	14.2%
24. Spencer Jones New York Yankees · vs RHP · 11 HR / 298 PA	13.5%	12.3%	13.4%	12.2%	12.4%	13.4%	13.4%	12.6%	12.7%
25. Miguel Vargas Chicago White Sox · vs LHP · 33 HR / 714 PA	13.4%	11.9%	13.4%	13.4%	13.2%	13.4%	13.4%	13.4%	13.6%
26. Andy Pages Los Angeles Dodgers · vs LHP · 23 HR / 574 PA	13.3%	12.8%	12.8%	12.5%	13.0%	12.8%	12.8%	12.8%	12.7%
27. Manny Machado San Diego Padres · vs RHP · 31 HR / 685 PA	13.3%	13.0%	13.4%	13.9%	13.7%	13.4%	13.4%	13.7%	13.8%
28. William Contreras Milwaukee Brewers · vs LHP · 18 HR / 640 PA	13.0%	12.8%	12.8%	12.6%	12.7%	12.8%	12.8%	12.7%	12.8%
29. George Lombard Jr. New York Yankees · vs RHP · 7 HR / 174 PA	12.6%	13.3%	12.8%	13.3%	13.2%	12.8%	12.8%	13.2%	12.6%
30. Jo Adell Cleveland Guardians · vs LHP · 23 HR / 661 PA	12.5%	11.7%	12.2%	12.7%	12.3%	12.2%	12.2%	12.3%	12.7%
31. Brice Turang Milwaukee Brewers · vs LHP · 21 HR / 680 PA	12.5%	12.8%	12.3%	12.4%	12.1%	12.3%	12.3%	11.8%	11.8%
32. Freddie Freeman Los Angeles Dodgers · vs LHP · 16 HR / 657 PA	12.3%	12.1%	12.0%	12.3%	11.8%	12.0%	12.0%	12.2%	12.0%
33. Kyle Tucker	11.9%	12.0%	12.0%	11.6%	11.9%	12.0%	12.0%	11.8%	11.6%

Hitter	Served (full season)	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression
Los Angeles Dodgers · vs LHP · 17 HR / 627 PA									
34. Cody Bellinger New York Yankees · vs RHP · 18 HR / 571 PA	11.8%	12.0%	11.7%	11.7%	11.9%	11.7%	11.7%	11.8%	11.7%
35. Michael Harris II Atlanta Braves · vs LHP · 27 HR / 634 PA	11.6%	11.7%	11.6%	12.1%	11.5%	11.6%	11.6%	11.8%	12.2%
36. Ty France San Diego Padres · vs RHP · 21 HR / 508 PA	11.0%	10.8%	11.3%	11.2%	11.3%	11.3%	11.3%	11.2%	11.2%
37. Christian Yelich Milwaukee Brewers · vs LHP · 12 HR / 500 PA	10.9%	11.1%	11.2%	10.7%	11.1%	11.2%	11.2%	11.4%	10.9%
38. Jazz Chisholm Jr. New York Yankees · vs RHP · 19 HR / 536 PA	10.7%	10.2%	10.8%	10.5%	10.5%	10.8%	10.8%	10.5%	10.5%
39. Jonny DeLuca Tampa Bay Rays · vs RHP · 11 HR / 370 PA	10.6%	11.3%	10.4%	10.8%	10.1%	10.4%	10.4%	10.4%	10.7%
40. Austin Wells New York Yankees · vs RHP · 14 HR / 396 PA	10.3%	10.0%	10.4%	10.0%	10.0%	10.4%	10.4%	10.2%	10.0%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.